Superpixels Using Pixel Neighborhood

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***Abstract*—Superpixel segmentation is an important preprocessing technique in image processing and computer vision. It reduces image complexity by grouping similar neighboring pixels into meaningful regions. Simple Linear Iterative Clustering (SLIC) is one of the most widely used superpixel algorithms because of its simplicity and computational efficiency. However, standard SLIC may suffer from over-segmentation, weak boundary preservation, and limited local context awareness. This paper presents an improved SLIC-based segmentation approach using pixel neighborhood information and adaptive region merging. The proposed method first applies SLIC to generate initial superpixels and then analyzes local neighborhood relationships to merge similar adjacent regions. Edge-aware and adaptive merging strategies are used to preserve important boundaries while reducing unnecessary regions. Experimental results on multiple images show that the proposed method produces more meaningful segmentation compared to standard SLIC.**

***Index Terms*—Superpixel, SLIC, Image Segmentation, Pixel Neighborhood, Adaptive Region Merging, Edge-Aware Segmentation**

I. Introduction

Image segmentation is a fundamental task in image processing and computer vision. The main goal of segmentation is to divide an image into meaningful and homogeneous regions so that further analysis becomes easier. It is widely used in object detection, medical imaging, autonomous driving, remote sensing, image compression, and scene understanding.

Traditional pixel-level image processing can be computationally expensive because every pixel is processed individually. For high-resolution images, this becomes inefficient and time-consuming. Superpixels solve this problem by grouping similar neighboring pixels into compact regions. Instead of working with millions of pixels, an algorithm can work with a smaller number of superpixels.

Among different superpixel methods, Simple Linear Iterative Clustering (SLIC) is popular because it is simple, fast, and effective. SLIC groups pixels based on color similarity and spatial proximity. However, standard SLIC also has some limitations. It may generate too many small regions, fail to preserve important boundaries, and may not fully consider local neighborhood information.

To address these limitations, this work proposes an improved segmentation approach based on SLIC, pixel neighborhood analysis, and adaptive region merging. The proposed method uses the initial SLIC output and then merges similar neighboring regions using local information. This helps reduce over-segmentation and improves visual quality.

II. Objectives The

main objectives of this project are:

- To apply SLIC superpixel segmentation on input images.
- To analyze the effect of pixel neighborhood information.
- To study the influence of parameters such as *n segments* and compactness.
- To propose an improved region merging method.
- To compare standard SLIC with the proposed method using visual and quantitative results.

III. Related Work

Superpixel segmentation has been widely studied in computer vision. Achanta et al. introduced the SLIC algorithm, which clusters pixels in a five-dimensional feature space using color and spatial coordinates. SLIC is computationally efficient and produces compact superpixels.

Graph-based segmentation methods also use relationships between neighboring pixels or regions. These approaches consider connectivity and similarity between nearby components. However, some graph-based methods can be computationally expensive.

Recent segmentation methods often combine superpixels with edge detection, clustering refinement, and region merging. These techniques improve segmentation quality by considering local image structure. In this project, we combine SLIC with pixel neighborhood analysis and adaptive region merging to improve the segmentation result.

IV. Background of SLIC

SLIC works by clustering pixels in a five-dimensional space:

$$[L_*, a_*, b_*, x, y]$$

where L_* $_*$ $_*$, a, b represent Lab color values and x, y represent spatial coordinates.

The approximate grid interval S is defined as:

$$S = \sqrt{\frac{N}{K}}$$

where N is the total number of pixels and K is the desired number of superpixels.

The distance measure in SLIC combines color distance and spatial distance:

$$D = \sqrt{\left(\frac{d_c}{m}\right)^2 + \left(\frac{d_s}{S}\right)^2}$$

where d_c is color distance, d_s is spatial distance, m is the compactness parameter, and S is the grid interval.

A key advantage of SLIC is that each cluster center searches only within a local $2S \times 2S$ neighborhood. This reduces computational cost compared to global search.

V. Proposed Methodology

The proposed method improves standard SLIC by adding adaptive region merging based on pixel neighborhood information.

A. Input Image

The input image is first loaded and converted into RGB format. If the image is grayscale, it is converted into a threechannel RGB image. The image is resized if necessary to reduce computation time.

B. Initial SLIC Segmentation

Standard SLIC is applied to generate initial superpixels. These superpixels form the base segmentation. Each region contains pixels that are similar in color and close in spatial position.

C. Pixel Neighborhood Analysis

After generating superpixels, neighboring regions are analyzed. Each superpixel is treated as a region, and adjacent regions are identified. This local relationship helps determine whether two regions should remain separate or be merged.

D. Edge-Aware Merging

An edge map is generated using the Sobel operator. Strong edges usually indicate object boundaries. Therefore, regions separated by strong edges should not be merged. This helps preserve important image structures.

E. Adaptive Thresholding

Instead of using a fixed merging threshold, the proposed method uses an adaptive threshold based on local variation. Smooth regions are more likely to be merged, while complex regions with strong variation are preserved.

F. Region Merging

Similar neighboring regions are merged based on color similarity, spatial relationship, intensity variation, and edge strength. This reduces unnecessary regions and produces a smoother segmentation result.

VI. Proposed Workflow

Fig. 1 shows the proposed method pipeline. The workflow starts from the input image, applies SLIC segmentation, analyzes pixel neighborhood information, performs adaptive region merging, and produces the final segmented output.

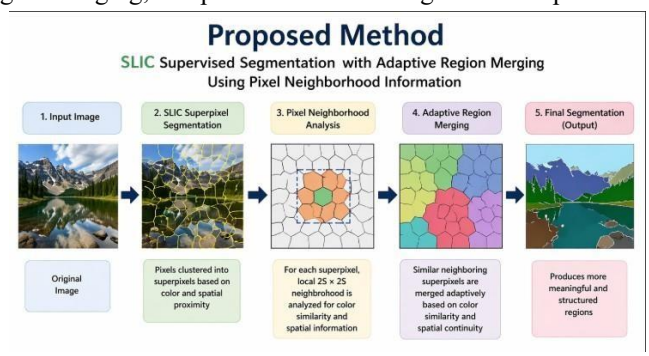


Fig. 1. Proposed method pipeline using SLIC segmentation and adaptive region merging.

The proposed workflow shows that the method does not depend only on basic SLIC segmentation. Instead, it uses additional neighborhood-based processing to improve the final result.

VII. Experimental Setup

The experiment was performed on five different images including human face, grass, background, object, and tree images. These images were selected to test the performance of the method on different visual structures. The following parameters were analyzed:

- Number of segments
- Compactness
- Region count
- Boundary ratio
- Execution time

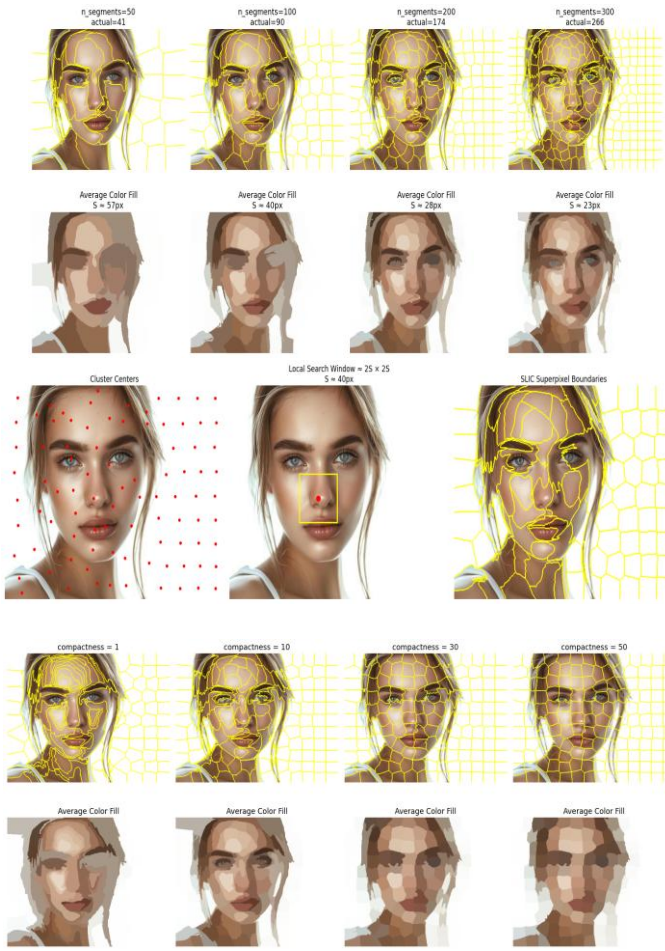
A. Effect of Number of Segments

The *n segments* parameter controls the approximate number of superpixels. A higher value produces more superpixels and finer segmentation. However, too many superpixels may cause over-segmentation.

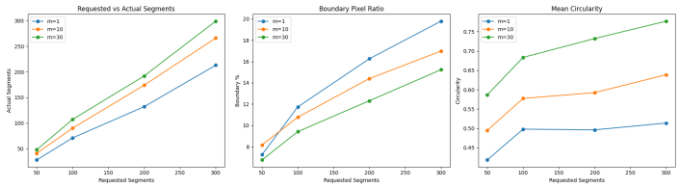
B. Effect of Compactness

The compactness parameter controls the balance between color similarity and spatial regularity. A low compactness value

follows boundaries more closely, while a high compactness value produces more regular shapes.



	requested_segments	compactness	actual_segments	avg_region_size_px	boundary_percent	circularity_mean	S_px	window_px
0	50	1	28	5812.50	7.247	0.4184	57	114
1	50	10	41	3969.51	8.160	0.4945	57	114
2	50	30	48	3390.62	6.750	0.5864	57	114
3	100	1	71	2292.25	11.735	0.4980	40	80
4	100	10	90	1808.33	10.768	0.5773	40	80
5	100	30	107	1521.03	9.411	0.6832	40	80
6	200	1	132	1232.95	16.260	0.4962	28	56
7	200	10	174	935.34	14.397	0.5923	28	56
8	200	30	192	847.66	12.313	0.7320	28	56
9	300	1	213	764.08	19.790	0.5137	23	46
10	300	10	266	611.84	16.976	0.6386	23	46
11	300	30	299	544.31	15.228	0.7769	23	46



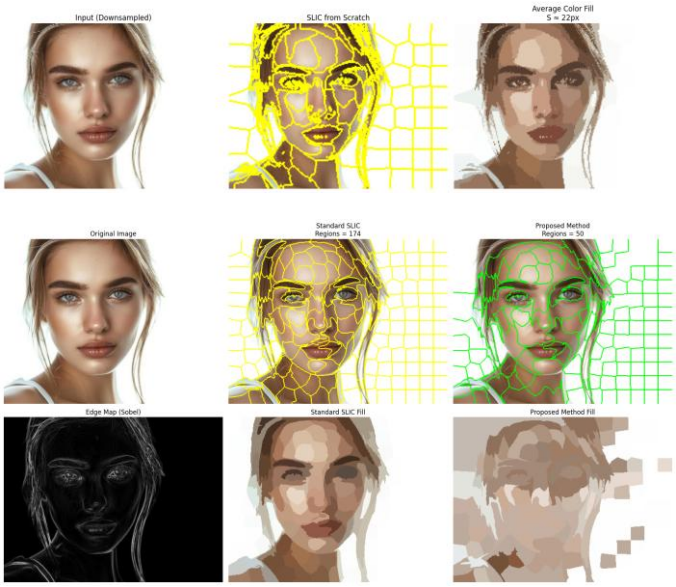
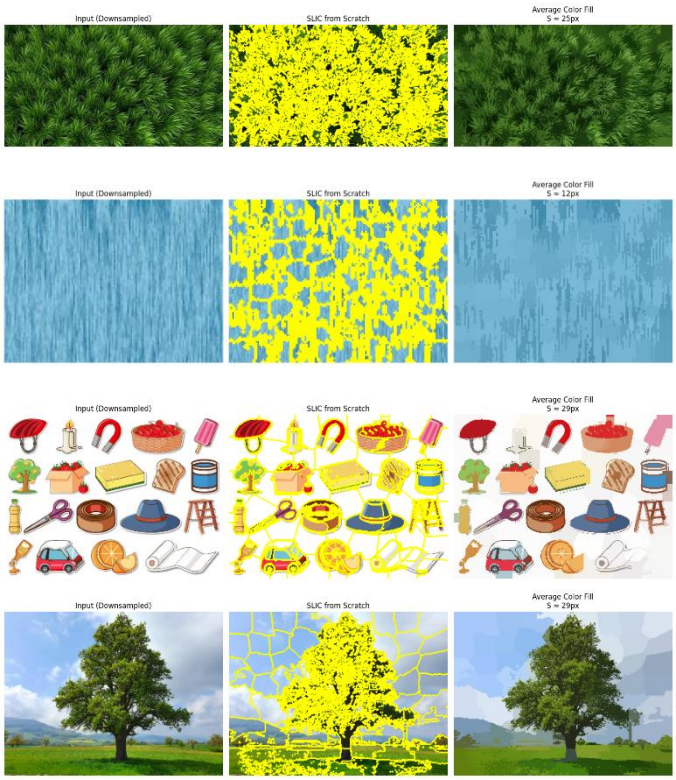


	image	method	regions	boundary_ratio	compactness	execution_time_sec
0	human face.jpg	Standard SLIC	174	0.1440	0.5923	0.3376
1	human face.jpg	Proposed Method	50	0.1173	0.4289	12.4977



VIII. Results and Discussion

The proposed method was compared with standard SLIC using multiple evaluation metrics. Fig. 2 shows the comparison between standard SLIC and the proposed method.

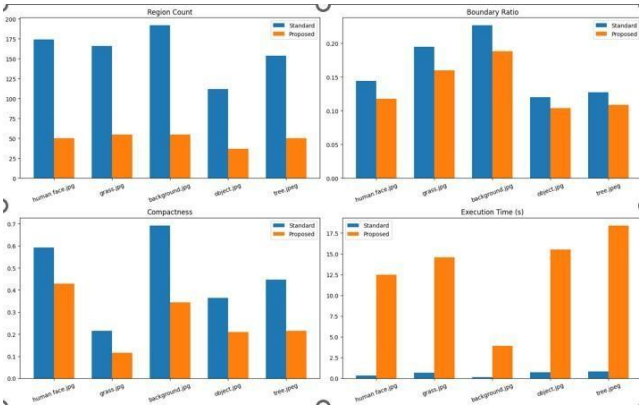


Fig. 2. Comparison between standard SLIC and the proposed method using region count, boundary ratio, compactness, and execution time.

From the comparison, it can be observed that the proposed method reduces the number of regions compared to standard SLIC. This indicates that unnecessary over-segmentation is reduced. The boundary ratio also decreases, showing that fewer unnecessary boundaries remain after adaptive merging.

The proposed method requires more execution time because it performs additional edge-aware and adaptive merging operations. However, this extra computation is acceptable because it improves segmentation quality. The method produces more meaningful and structured regions.

IX. Evaluation Metrics

The following metrics were used to evaluate the segmentation result.

A. Region Count

Region count represents the total number of segmented regions. A very high region count may indicate oversegmentation. The proposed method reduces region count by merging similar neighboring superpixels.

B. Boundary Ratio

Boundary ratio measures the proportion of boundary pixels in the image. Lower unnecessary boundary ratio indicates smoother segmentation.

C. Compactness

Compactness measures the regularity of segmented regions. It helps analyze whether regions are smooth and structured.

D. Execution Time

Execution time measures computational cost. The proposed method takes more time than standard SLIC because it performs extra merging operations. However, it provides better segmentation quality.

X. Advantages of the Proposed Method

The proposed method has several advantages:

- Reduces over-segmentation.
- Preserves important boundaries.
- Uses local pixel neighborhood information.
- Produces more meaningful regions.
- Works on different image types.

XI. Limitations

Although the proposed method improves segmentation quality, it also has some limitations:

- Execution time is higher than standard SLIC.
- Parameters need tuning for different images.
- Highly textured images may still be difficult to segment perfectly.

XII. Applications

Superpixel segmentation can be used in many computer vision applications, such as:

- Object detection
- Medical image segmentation
- Image compression
- Scene understanding
- Autonomous driving
- Remote sensing

XIII. Conclusion

This paper presents an improved SLIC-based superpixel segmentation method using pixel neighborhood information and adaptive region merging. Standard SLIC is first used to generate initial superpixels. Then, neighboring regions are analyzed and similar regions are merged using edge-aware and adaptive strategies.

The experimental results show that the proposed method reduces unnecessary segmentation and produces more meaningful regions. Although execution time increases due to additional processing, the segmentation quality improves. Therefore, the proposed method provides a better balance between segmentation quality and computational efficiency.

XIV. Future Work

Future improvements may include:

- Automatic parameter tuning.
- Deep learning-based region refinement.
- Real-time implementation.
- Testing on larger benchmark datasets.
- Application in medical and remote sensing images.

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